Convolutional Neural Networks

Agenda

- Convolutional Neural Networks Quiz
- Convolutional Filter
- Pooling
- Data Augmentation
- Spatial Dropout
- Transfer Learning

Let's begin the discussion by answering a few questions

Convolutional Neural Networks Quiz

What is the output of convolving a 2x2 filter of all ones with any 2x2 input matrix of real values?

A The product of all elements in the input matrix.

B The mean of all elements in the input matrix

C The sum of all elements in the input matrix.

D The maximum value of all elements in the input matrix.

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Convolution Operation

Refers to the process of applying a filter to the input data by computing the element wise product between the filter and the local regions of the input

a	b
c	d

*

1	1
1	1

$= a * 1 + b * 1 + c * 1 + d * 1$

$= a + b + c + d$

Thus, we get the sum of all elements of the input matrix

Convolutional Neural Networks Quiz

What is the total number of learnable parameters in a Max Pooling layer ?

A. Depends on the kernel size.

B. 1

C. Depends on the size of the input feature map

D. 0

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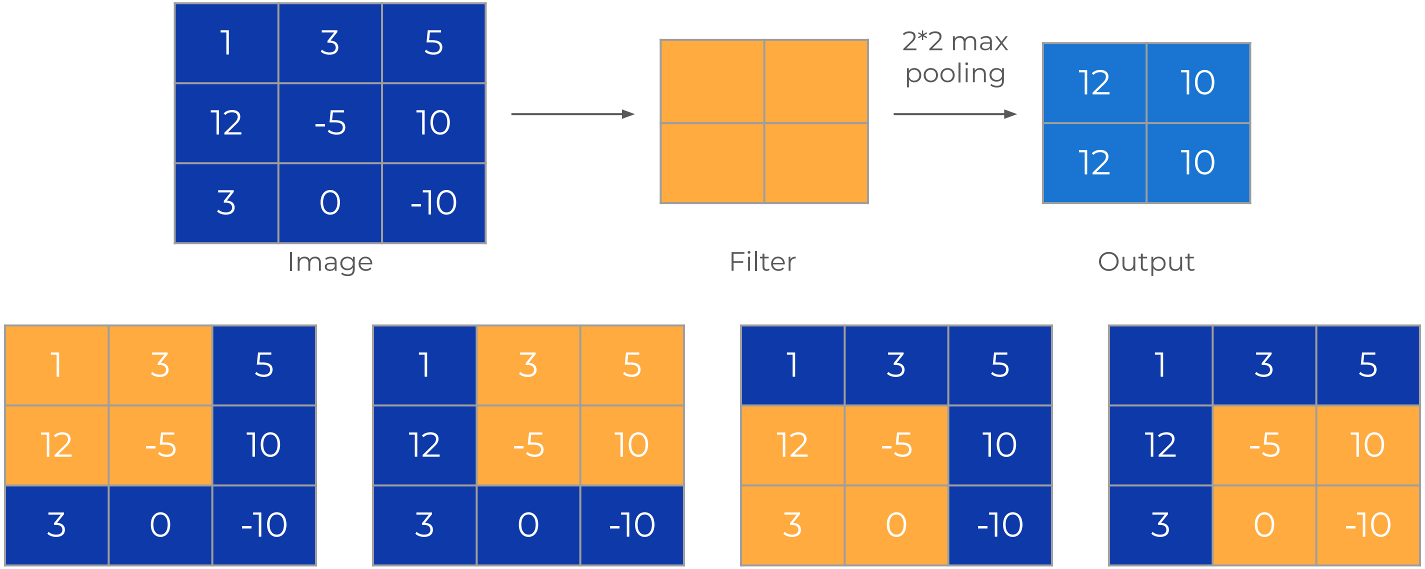
C Depends on the size of the input feature map

D 0

Pooling

Decreases the spatial dimensions of feature maps

Reduction in size aids in managing computational complexity, facilitating faster training

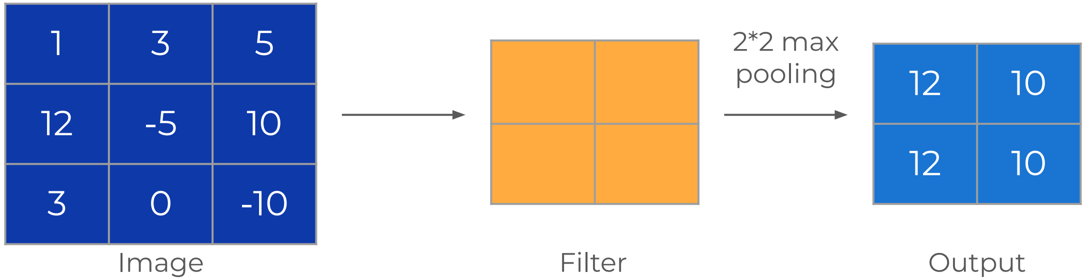


Taking the maximum value in each of the regions.

Max Pooling

Operates by selecting the maximum value within each pooling region, without any trainable weights or biases.

As we are performing an operation to find the maximum value, there are **no learnable parameters**



Convolutional Neural Networks Quiz

Which of the following statements best describes the role of data augmentation in Convolutional Neural Networks (CNNs)?

A. Helps to increase the complexity of the model architecture.

B. Assists in reducing overfitting by introducing artificial variations to the training dataset.

C. Primarily focuses on reducing the computational resources required for training CNNs.

D. Removes some of the images from the training dataset.

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Data Augmentation

Introduces artificial variations to training data

Helps reduce overfitting by exposing the model to diverse examples

Enhances robustness and generalization of Convolutional Neural Networks

Original Image

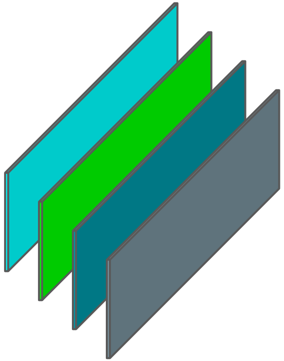
After Rotation

Changing color

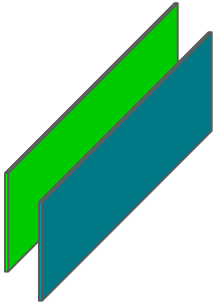
Reducing the brightness

Convolutional Neural Networks Quiz

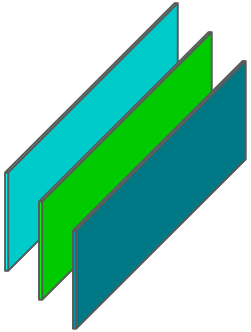
A spatial dropout is applied to four feature maps with a rate of 0.5. Which of the following best represents the scenario?



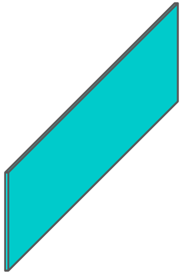
A



B



C



D

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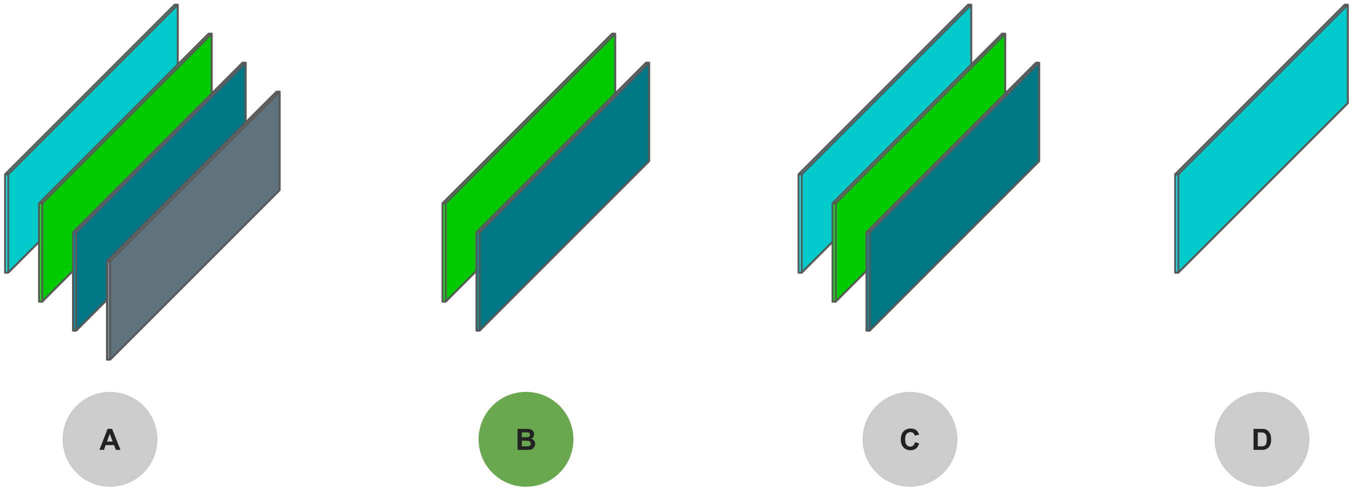
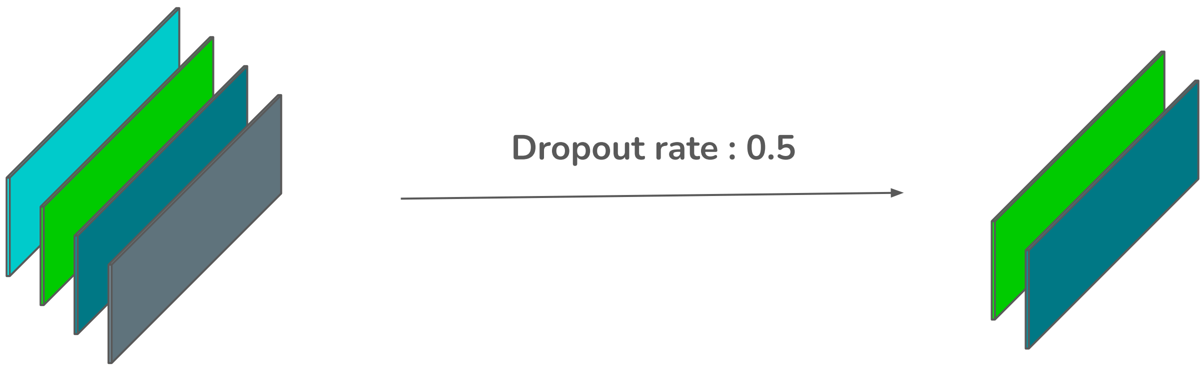


Spatial Dropout

Randomly sets entire feature maps to zero based on the rate during training in CNNs to prevent overfitting

Encourages the learning of robust and invariant features while acting as an ensemble technique, increasing the chances of generalized performance



Convolutional Neural Networks Quiz

Consider a VGG-16 model with the following architecture. What are the possible input dimensions (shape) that this model can accept?

Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0

A (128,128,3)

B (64,64,3)

C (53,53,3)

D (18,18,3)

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CNNs adoption to varying image sizes

If we closely observe the architecture, each max-pooling layer halves the input dimensions.

There are total of 5 such layers. (224 -> 112 -> 56 -> 28 -> 14 -> 7)

Option A	128	64	32	16	8	4
Option B	64	32	16	8	4	2
Option C	53	26	13	6	3	1
Option D	18	9	4	2	1	✖

If the input shape is (18,18,3), then with five max-pooling layers, it is not possible for this architecture to produce a output at all.

Convolutional Neural Networks Quiz

Which statement best explains the purpose of freezing layers during transfer learning in neural networks?

A. Freezing layers prevents the model from learning any new information during training.

B. Freezing layers allows the model to adapt to the new task more quickly by reusing pre-learned features.

C. Freezing layers ensures that the model's weights are always zero

D. Freezing layers enables the model to automatically adjust the learning rate for the new task.

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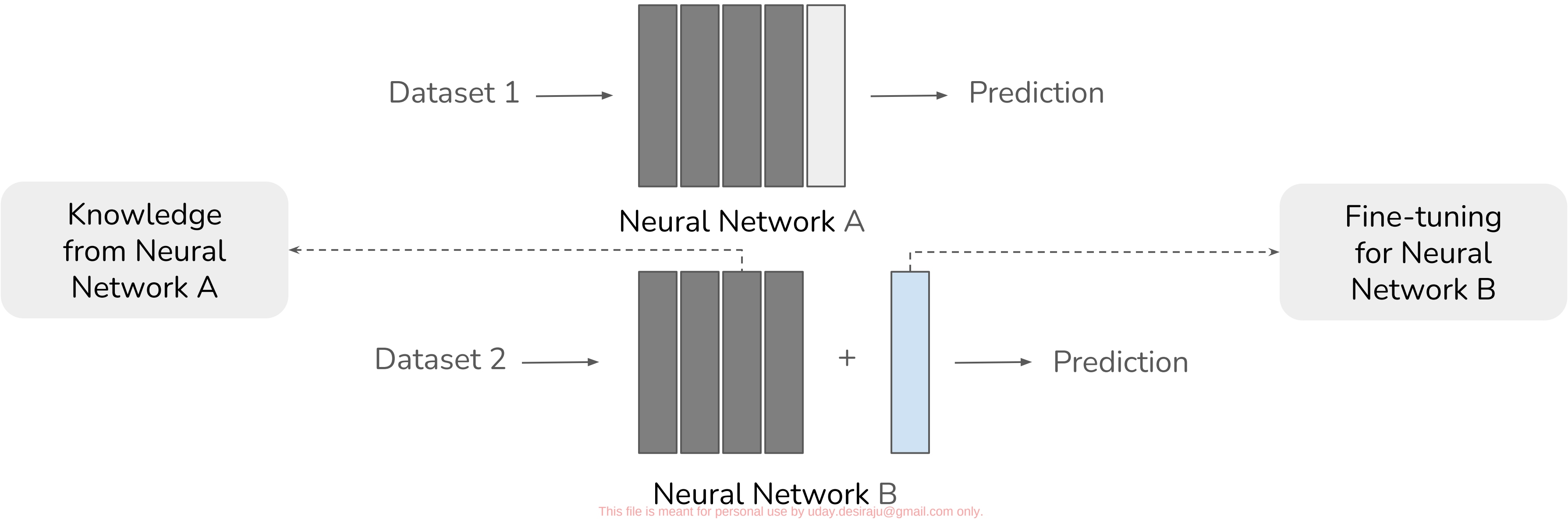
Transfer Learning

Applies knowledge from one task to improve performance on another

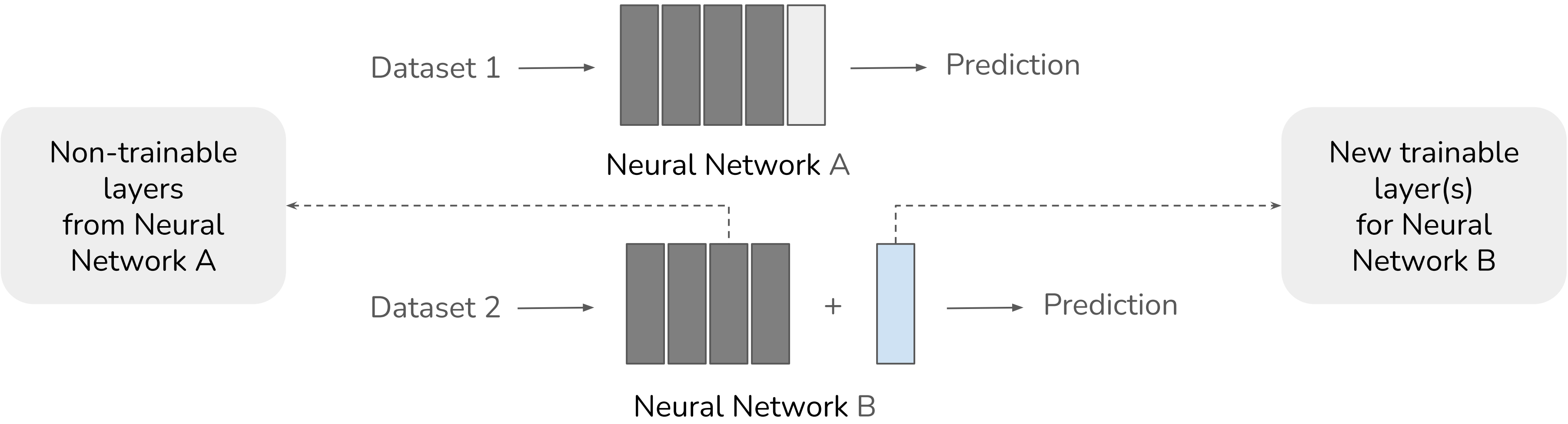
Fine-tuning pre-trained models adapts learned features to new tasks efficiently



Transfer Learning

Freezing layers retains pre-learned features, preventing them from being modified during training.

Gradients for frozen weights are ignored. This speeds up convergence and ensures that the model focuses on learning task-specific information.



Great Learning

Power Ahead!